where $K$ is the restoring couple per unit twist or torsional constant of the spring.

At equilibrium, the deflection couple is equal to the restoring couple. Therefore by comparing equation (3.69) and (3.70), we get

$$NABI = K\,\theta$$

$$\Rightarrow I = \frac{K}{NAB}\theta \qquad (3.71)$$

(or) $I = G\,\theta$

where, $G = \frac{K}{NAB}$ is called galvanometer constant or current reduction factor of the galvanometer.

Since, suspended moving coil galvanometer is very sensitive, we have to handle with high care while doing experiments. Most of the galvanometer we use are pointer type moving coil galvanometer.

**Figure of merit of a galvanometer**

***It is defined as the current which produces a deflection of one scale division in the galvanometer.***

**Sensitivity of a galvanometer**

The galvanometer is said to be sensitive if it shows large scale deflection even though a small current is passed through it or a small voltage is applied across it.

**Current sensitivity:** ***It is defined as the deflection produced per unit current flowing through it.***

$$I_S = \frac{\theta}{I} = \frac{NAB}{K} \Rightarrow I_S = \frac{1}{G} \qquad (3.72)$$

The current sensitivity of a galvanometer can be increased

(a) by increasing

(1) the number of turns N

(2) the magnetic induction B

(3) the area of the coil A

(b) by decreasing

the couple per unit twist of the suspension wire k. Phosphor - bronze wire is used as the suspension wire because the couple per unit twist is very small.

**Voltage sensitivity:** ***It is defined as the deflection produced per unit voltage applied across it.***

$$V_S = \frac{\theta}{V}$$

$$V_S = \frac{\theta}{IR_g} = \frac{NAB}{K R_g} \Rightarrow V_S = \frac{1}{GR_g} = \frac{I_S}{R_g} \qquad (3.73)$$

where $R_g$ is the resistance of galvanometer.

## EXAMPLE 3.29

The coil of a moving coil galvanometer has 5 turns and each turn has an effective area of $2 \times 10^{-2}$ m$^2$. It is suspended in a magnetic field whose strength is $4 \times 10^{-2}$ Wb m$^{-2}$. If the torsional constant $K$ of the suspension fibre is $4 \times 10^{-9}$ N m deg$^{-1}$.

(a) Find its current sensitivity in degree per micro - ampere.

(b) Calculate the voltage sensitivity of the galvanometer for it to have full scale deflection of 50 divisions for 25 mV.

(c) Compute the resistance of the galvanometer.

### Solution

The number of turns of the coil is 5 turns

The area of each coil is $2 \times 10^{-2}$ m$^2$

Strength of the magnetic field is $4 \times 10^{-2}$ Wb m$^{-2}$

Torsional constant is $4 \times 10^{-9}$ N m deg$^{-1}$

(a) Current sensitivity

$$I_S = \frac{NAB}{K} = \frac{5 \times 2 \times 10^{-2} \times 4 \times 10^{-2}}{4 \times 10^{-9}}$$

$= 10^6$ divisions per ampere



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